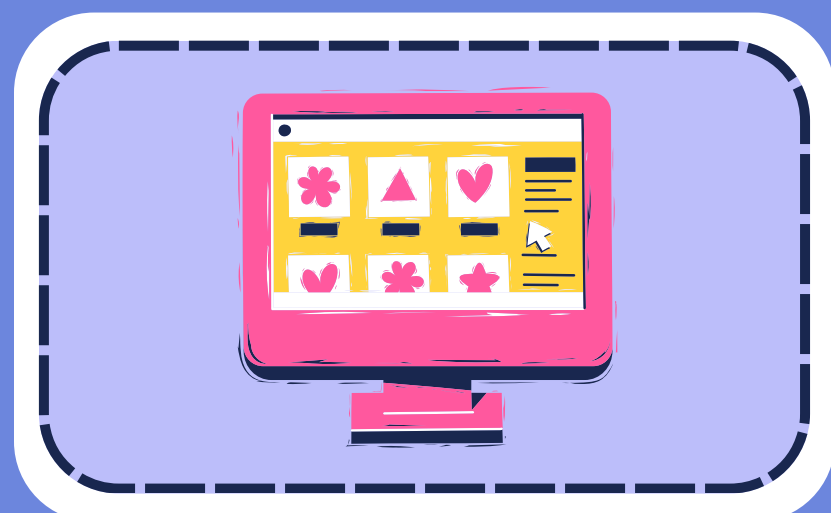
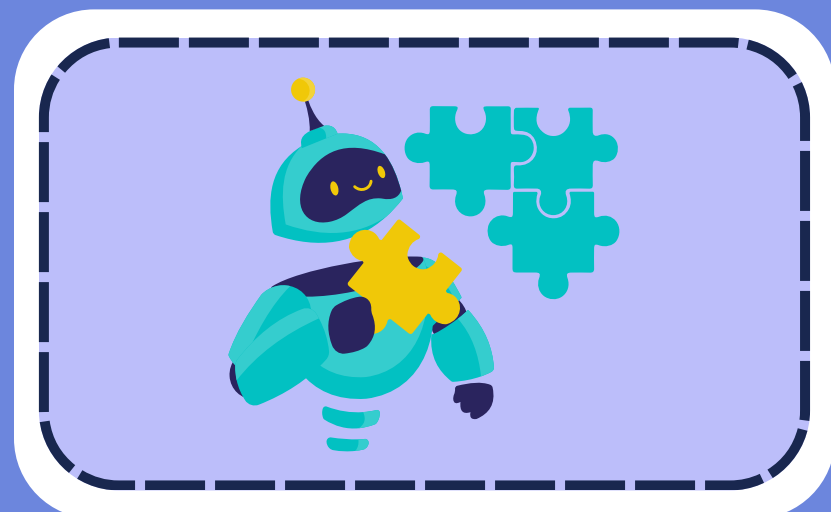
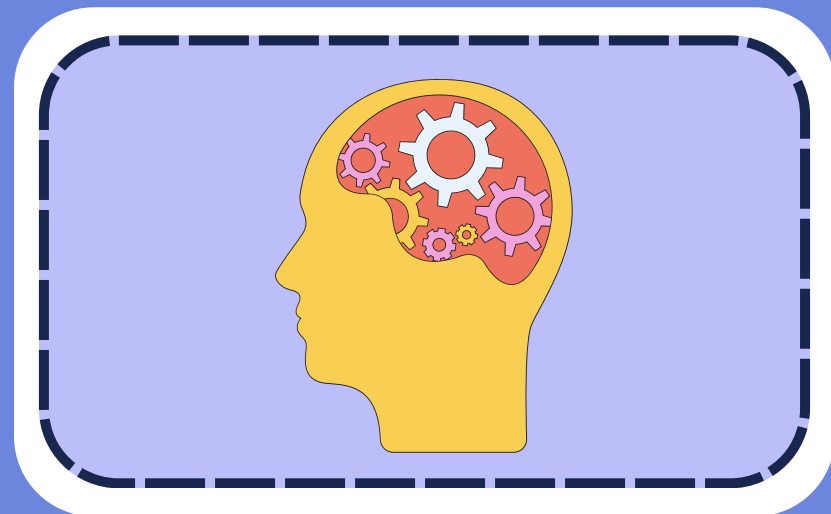




BYTE SIZED BUILDERS

**Building Bridges Between
Robotics & AI**





Cyber Survey for Future Scientists

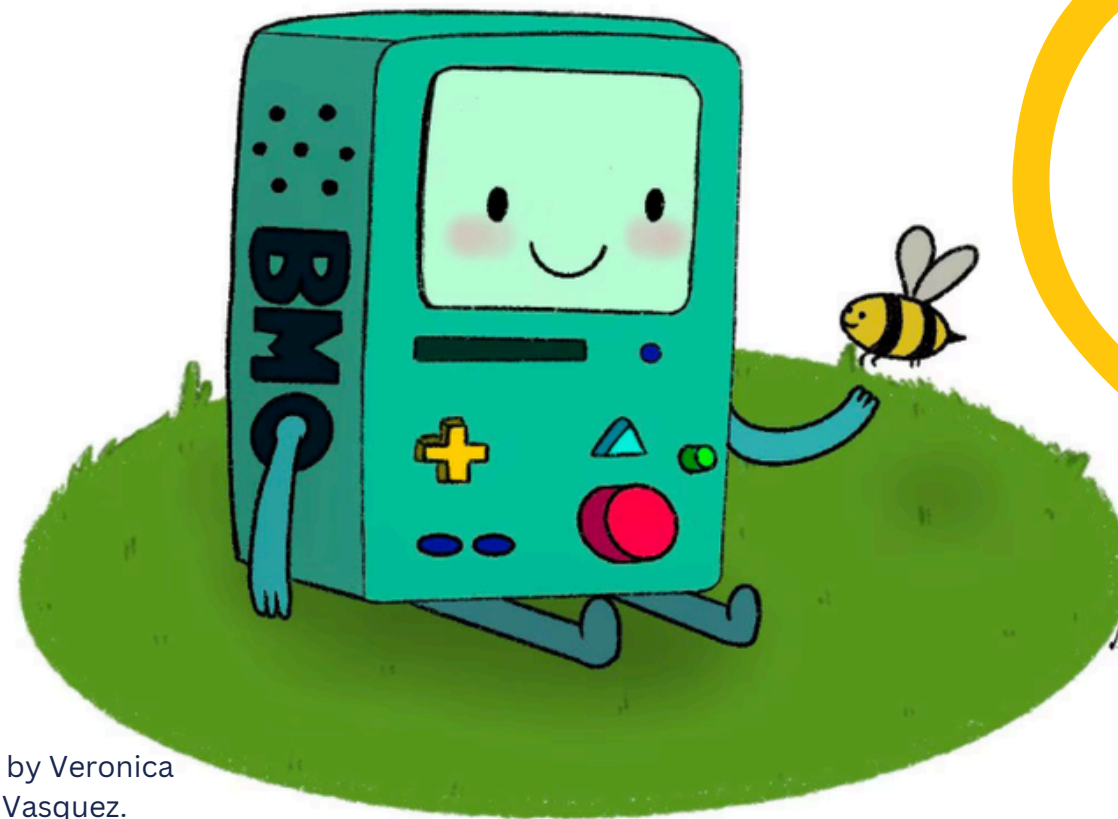
[https://kentstate.az1.qualtrics.com
/jfe/form/SV_1LIHadRPndGXn5Y](https://kentstate.az1.qualtrics.com/jfe/form/SV_1LIHadRPndGXn5Y)





Where have you seen AI?

Is BMO from Adventure Time AI?



Art by Veronica Vasquez.
See References.

Artificial Intelligence:
a computer program or app that can perform tasks that typically require human intelligence.

AI Chatbot:
a computer program or app that uses artificial intelligence technology to respond to what you say in real time.



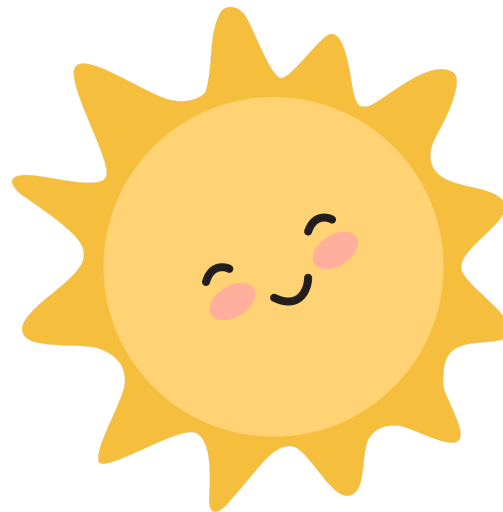
Oh, the places AI will go!

Healthcare



AI is used in healthcare to make quicker and more accurate diagnosis! AI can quickly detect cancer in medical images. It reduces medical errors and saves lives.

Weather Forecasting



AI helps weather experts study a ton of data to find sneaky weather patterns that we might miss! And because of this, we can make weather forecasts even more accurate!

Space Travel



CIMON 2 is a friendly floating robot in space that helps astronauts with tasks and keeps them company! It can get a little lonely up there.





Today we are AI Explorers!

- Explore the fun and fantastic frontier of AI to robotics.
- Join us for some fun games, cool prizes, and maybe even pick up some new knowledge!
- And to top it all off, we'll introduce you to our awesome robot friend!

But First...

What have you been learning in Robotics club?



HOW DO I MAKE A JELLY SANDWICH?

1

Grab a flash card and a marker.

2

Imagine you're a chef! Write down the steps to make a jelly and fluff sandwich.

3

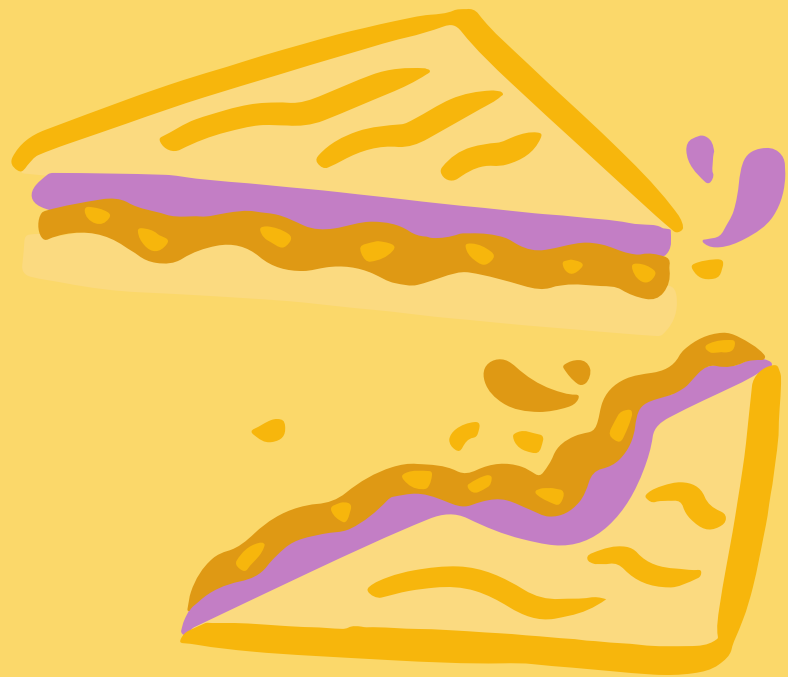
Pass your flashcard and marker forward.

4

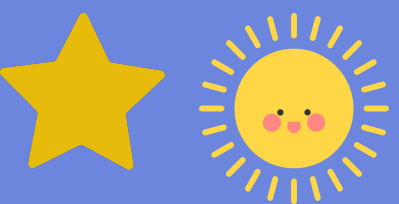
Let's pick a flashcard and make a sandwich!



Hmm That didn't go quite as planned...



- Chatbots don't have common sense like us, so they need super clear instructions to know what to do!
- It's like when you're at a fork in the road, and you have to decide which way to go. Chatbots use big decision trees to help them choose what to do next.
- Imagine making a yummy jelly sandwich. First, you unscrew the jelly lid, then you dip the butter knife into the jelly and spread it on a piece of bread. Finally, you put the two pieces of bread together, and ta-da! You've got yourself a delicious jelly sandwich!





Chatbots: Who or What is behind the screen?



AI CONTENT OR HUMAN CONTENT? CAN YOU TELL THE DIFFERENCE?

1

**Let's look
through a
series of
images.**

2

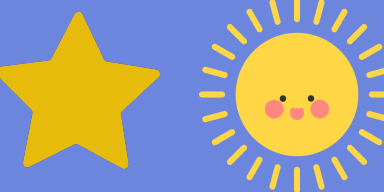
**Can you tell
which
images are
AI and
which are
Human
made?**

3

**We'll take a
vote on
each image.**

4

**Why did you
choose your
answer?
What clues
did you
find?**



Human or AI?



Human or AI?



Human or AI?



Human or AI?

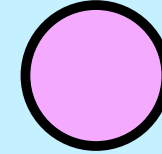


Human or AI?

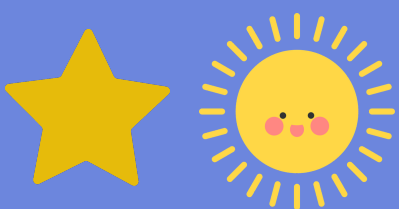


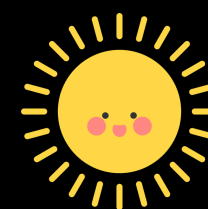


AI VS HUMANS IN GAMES



- Did you know that special AI bots can learn to play video games, just like us?
- In 2019, these clever bots even played against human players in a game called DOTA, and guess what? They won!
- Bots can also be programmed to find the shortest path to complete a game. This is different from an AI bot who learns to play the game through trial and error.







You can recognize AI! Can AI recognize you?

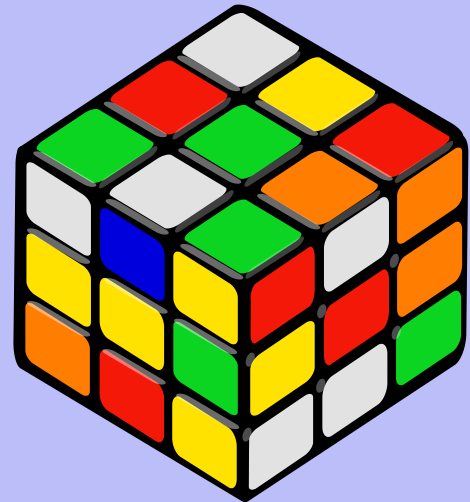
Let's Play Quick Draw!



Quick Draw, developed by Google, employs advanced neural networks to guess what you're drawing! With each round, it learns and improves its predictions.

Neural Networks are like smart machines inspired by how our brains work! They're really good at understanding tricky stuff and figuring out answers.





ROBOTS WITH SUPERPOWERS

How do we humans learn about our environment? Through our senses! Robots can learn about their environment through artificial senses. The sense of sight can help a robot solve a Rubik's cube.

Computer Vision

Robots can learn and do things just like us by watching and copying what humans do. But how do they see? Well, they use something called computer vision! It's like giving the robot eyes. They look at lots and lots of pictures to learn and recognize patterns.

Robots have Many Senses

People have five senses: touch, sight, hearing, smell, and taste. These senses help us know what's happening around us. Robots can have their own special senses too! They might have sensors to feel temperature, touch things, sense vibrations, and more!



My Robot Friend Needs Help!

My robot friend, Zivko, needs some repairs, but I need your help to do it! Without their repairs, Zivko can't go on new adventures!

Infrared Sensor.

Zivko uses a special infrared sensor to check what's around them, kind of like how we use our eyes. This sensor helps Zivko avoid bumping into things. Even though we can't see infrared, we can feel it as warmth, like when we stand in the sun!

Explore and Follow

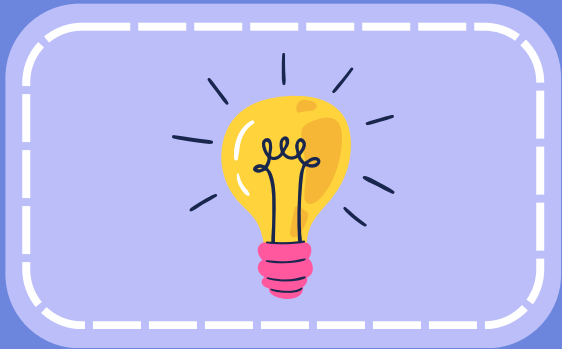
Zivko loves exploring! They have two modes: 'Explore' and 'Follow.' They use special technology and sensors to see around them and avoid things in their way. And if Zivko gets lost, I can make them follow me instead!

Let's Build!

Our job is to fix up Zivko and try out their two different modes.

First, we'll read the instructions to figure out what part of Zivko needs fixing!





References

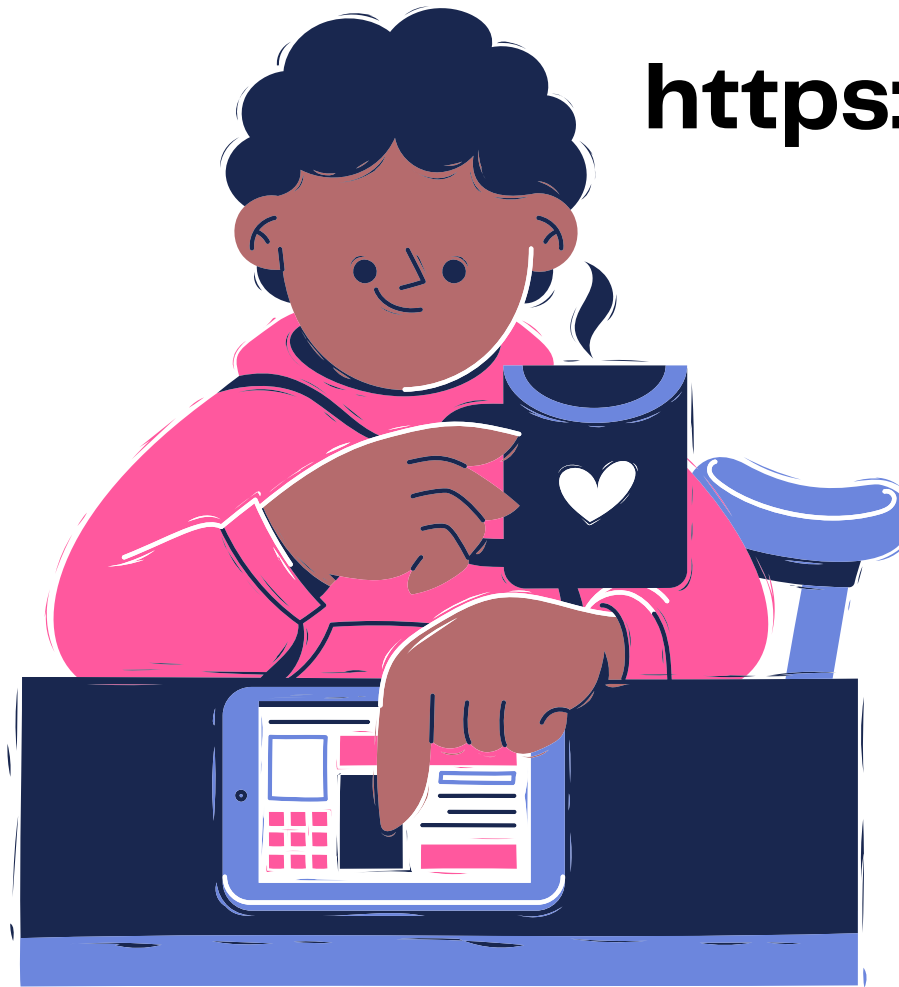
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- Note: Human VS AI game slides 10, 12, and 13 generated by me using NightCafe AI Art Generator.



Cyber Survey for Future Scientists

https://kent.qualtrics.com/jfe/form/SV_2t19ZZ6OiACuaaO



**Press
these keys
while on
Present
mode!**



for blur



for confetti



for a drumroll



for mic drop



for bubbles



for quiet



for unveil



Any number from
0-9 for a timer